PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM TWO MOCK EXAMINATION
Code: 041 **BASIC MATHEMATICS**
TIME: 2:30 Hours **May 2024**

INSTRUCTIONS:

1. This paper consists of ten (10) compulsory questions.
2. Show clearly all the working and answers in the space provided.
3. Mathematical tables and graph papers may be used.
4. Write your name on every page of your answer sheet.

QUESTIONS:

1. (a) Find the sum of the prime numbers between 70 and 90.

(b)At Mtakuja Secondary school $\frac{5}{7}$ of students are boys. If there 120 girls in the school, find the number of students in the school.

(c) In form two class, 5% of the students can play football, $\frac{1}{4}$ can play volleyball, 0.1 can play Basketball and $\frac{3}{5}$can play tennis. Arrange these numbers in descending order.

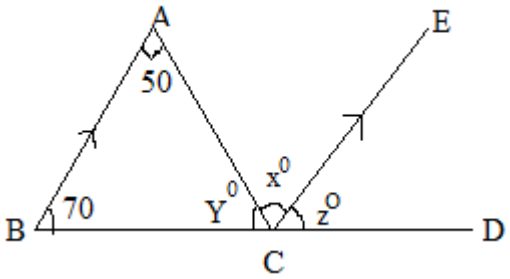
2. (a) Change thirty minutes past six in the evening using
   (i) 12 – hours clock.
   (ii) 24- - hours clock.

(b) Convert the following into Kilograms.

(i)4800 grams.
(ii)0.05 tones.

(c) Anna bought 3 bottles of juice of 350 milliliters each and John bought 1 bottle of juice of 1Littre. Who had more juice to drink? What is the difference in capacity of their juice?

3. (a) Find the values of x, y and z.



(b) The area of a trapezium whose height is 8cm, is 236 cm^2 If one of its parallel sides is longer than the other by 4 cm, find the lengths of the two parallel sides.

4. (a) The sum of the number of boys and girls in the class is 36. If twice the number of girls exceeds the number of boys by 12, find the number of girls and that of boys in the class.

(b) Solve the following by the method of completing the square.

$2y^2 + 5y + 3 = 0$.

5. (a) Find the ratio A : B : C, if A : B = 3: 5 and A : C = 6: 7

(b) A machine costing 180,000/= Tanzanian shillings is sold at a profit of 40%. What is its selling price?

6. (a) Find the solution of $|2X + 1| > 3$, and show it on a number line.

(b) Find x – intercept and y – intercept of the line passing through the following pair of points (-2, 1) and (4, -1).

7. (a) Rationalize the denominator and simplify $\frac{5\sqrt{3}}{3\sqrt{2}+\sqrt{3}}$.

(b) Given that $T = 2\pi\sqrt{\frac{X}{2}}$ make x the subject of the formula.

(c) Solve for x and y in $(2^X)(5^Y) = 100$

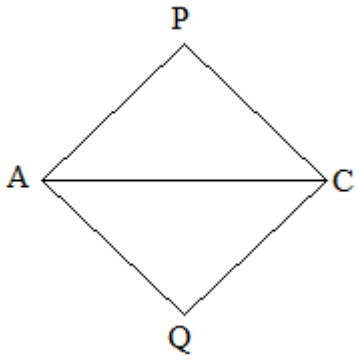
8. (a) Find the value of y if $\log_y 81 - \log_2 32 = -1$

(b) Find the value of $\log \sqrt[3]{x}$ if $\log x = \bar{8}.0524$

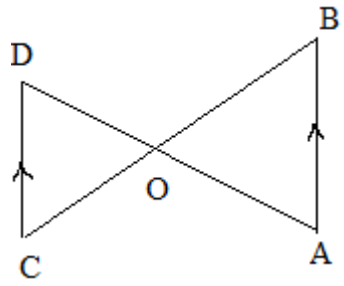
(c) Evaluate; $\log\left(\frac{764\times36.56}{\sqrt{3845}}\right)$

$\log 7.64 = 0.8831$
Given that; $\log 3.656 = 0.5630$
$\log 3.845 = 0.5849.$

9. (a) In the following figure, $\overline{AC}$ bisects $P\hat{A}Q$ and $P\hat{C}Q$ prove that $\overline{AP} = AQ$.



(b) In the following figure $\overline{CD} // \overline{AB}$, prove that; Δ ABO $\sim \Delta$ CDO.



10. A rectangular field of 80M long is to be fenced with a wire. If the diagonal of the field is 100M long find;
    (a) Width of the field.
    (b) Length of the wire needed to fence the field.